

7 People vary in which molecules are present on the surface of their red blood cells. For example, in the ABO blood group system, people may be categorised into one of four blood groups: A, B, AB and O.

A single gene, *I*, with three alleles, determines these blood groups. Gene *I* is an example of a gene with multiple alleles.

- Alleles **I^A** and **I^B** are codominant.
- Allele **I^o** is recessive to alleles **I^A** and **I^B**.

(a) Explain what is meant by codominant and recessive.

codominant

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recessive

.....

[3]

(b) Table 7.1 shows a cross between a person with blood group A and a person with blood group B.

Complete Table 7.1.

Table 7.1

parental phenotype (blood group)	A		B	
parental genotype	I^A I^o		I^B I^o	
gametes				
offspring genotype				
offspring phenotype (blood group)				

[3]

- (c) Haemophilia is a blood condition where bleeding continues for a long period of time after a blood vessel has been damaged.

A person with haemophilia has a mutation in a gene on the X chromosome, which results in the lack of a blood clotting factor.

Explain why a man who has haemophilia cannot pass the condition to his male offspring.

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..... [3]

[Total: 9]